Which Steam tag pair should an indie studio build next? Puzzle × Story Rich

MADT 7202 Mid-Term · Panjapol Ampornratana 6810424009 · Repo: github.com/hhej/steam-tag-whitespace

1 The decision

The creative director of an indie studio decides which two-tag game concept to build next. If no pair had passed the test below, they would pick the best-performing crowded pair, Management × Strategy.

2 Why a network

Tag counts show which tags are common. They cannot show which two tags sit in separate parts of the tag network yet do well when combined; that needs each tag's neighbours.

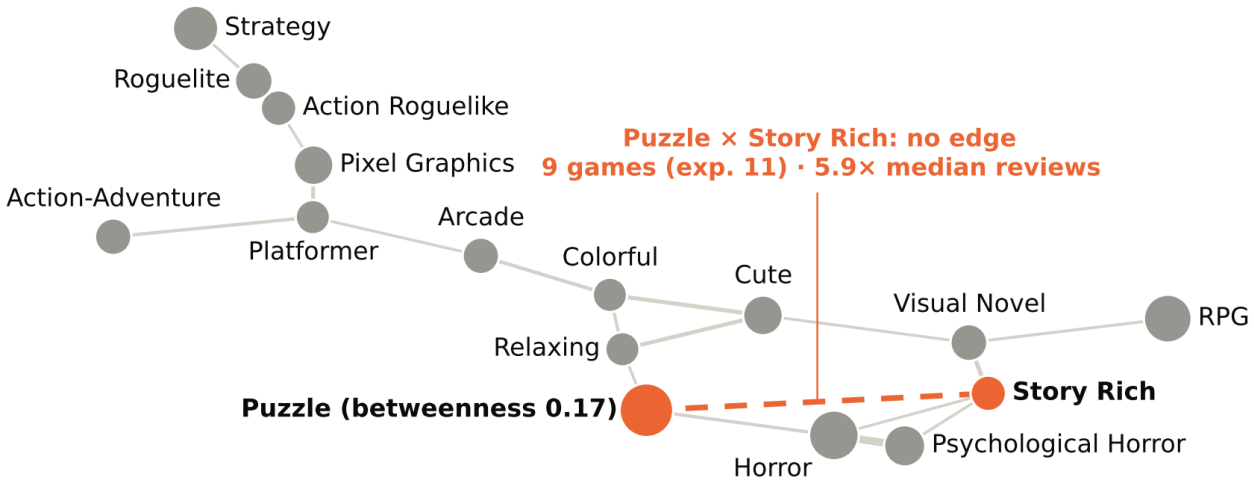
3 Data

Source	S1 FronkonGames Steam Games Dataset (Hugging Face). S2 Steam store pages and review API.
How obtained	S1 downloaded. S2 scraped 2026-10-14, 1 request per 1.5 s.
Legal basis	S1 MIT licence. Steam robots.txt allows /app/ and /appreviews/. Game-level data only, no personal data.
Period	Indie games released 1 Jan 2023 – 31 Dec 2024.
Volume	9,094 games with ≥10 reviews → random 1,000 (seed 7202); 996 analysed.
Cross-check	S1 vs S2 top-5 tags: Jaccard 0.87; review counts ρ = 0.98. SteamSpy dropped (copy of S1).
Bias	The ≥10-review floor favours games that sold. Reviews count buyers of existing games, not unmet demand.

4 Graph construction

Node	A tag.
Edge	Both tags are in the same game's top 5 tags by votes.
Weight	NPMI. With raw counts, 9 of the top-10 edges involve the 5 most common tags; with NPMI, 0.
Window	Top 5 tags, because the store page shows the top tags. With all 20 tags, 72 games carry the pair and its lift drops from 5.9× to 1.9×.
Excluded	"Indie" (the seed, on every game); tags on >25% of games (Singleplayer, Adventure, Casual, Action, 2D, 3D, Atmospheric, Exploration, Simulation, First-Person); release-model and adult tags.
Pruning	Tag on ≥15 games; edge NPMI ≥ 0.10 and ≥5 shared games.
Size	391 tags → 98 nodes, 249 edges. The figure shows 16.

5 The picture



Node size = games with the tag · edge width = NPMI · dashed = recommended pair (no edge).

6 Findings

- Puzzle has the highest betweenness (0.17) and links 14 tags from different parts of the network.
- **Puzzle × Story Rich**: 9 games (expected 11), no edge between them, neighbour overlap 0.10. Their median game has 302 reviews, 5.9× the cohort median of 51 (permutation p = 0.021).
- Management × Strategy has the highest lift (13.0×) but already has 2.4× more games than expected. Puzzle × Story Rich stays #1 at all 8 pruning cutoffs and on S1 tags. Lift is above 1 in 99/100 bootstraps, but the pair is #1 in only 3/100 because only 9 games support it.

7 Recommended action

- Pitch Puzzle × Story Rich at the next greenlight meeting, with Management × Strategy as plan B.
- Build a prototype and open a Steam Coming Soon page with Puzzle and Story Rich as its first two tags.
- Re-run this analysis on 2025 releases before full production. If the pair drops out of the top 3, switch to plan B.

8 What would change my mind

Only 9 games support the pair, so one hit game could explain the lift. If the pair drops out of the top 3 on 2025 releases, I switch to Management × Strategy.